Practical 10

Aim: Implementing a Virtual Memory System with Demand Paging where pages are loaded into memory only when needed, optimizing memory usage and minimizing page faults.

Code:
```
#include <stdio.h>

int main() {
   int frames, pages;

   printf("Enter number of frames: ");
   scanf("%d", &frames);

   printf("Enter number of pages: ");
   scanf("%d", &pages);

   int ref[pages];
   printf("Enter page reference string: ");
   for(int i = 0; i < pages; i++) {
      scanf("%d", &ref[i]);
   }

   int frame[frames];
   for(int i = 0; i < frames; i++) {
      frame[i] = -1; // empty
   }

   int pageFaults = 0, index = 0;

   for(int i = 0; i < pages; i++) {
      int found = 0;
```

```
        // Check if page already in frame
        for(int j = 0; j < frames; j++) {
            if(frame[j] == ref[i]) {
                found = 1;
                break;
            }
        }

        if(found == 0) {
            // Page Fault
            frame[index] = ref[i];
            index = (index + 1) % frames;
            pageFaults++;
        }

        // Display frame status
        printf("\nFrames: ");
        for(int j = 0; j < frames; j++) {
            printf("%d ", frame[j]);
        }
    }

    printf("\nTotal Page Faults = %d\n", pageFaults);

    return 0;
}
```

Output

```
Enter number of frames: 3
Enter number of pages: 7
Enter page reference string: 1 2 3 4 1 2 5

Frames: 1 -1 -1
Frames: 1 2 -1
Frames: 1 2 3
Frames: 4 2 3
Frames: 4 1 3
Frames: 4 1 2
Frames: 5 1 2
Total Page Faults = 7


=== Code Execution Successful ===
```